

ALGORITHM

Spring 2026, Final Exam, June 10

5:00 – 8:00 pm

1. (25%) Please answer the following questions:

- Illustrate the operation of RADIX-SORT on the following numbers: 457, 654, 578, 254, 957, 499, 558, 911, 731, 536.
- In the algorithm SELECT, the input elements are divided into groups of 5. Will the algorithm work in linear time if they are divided into groups of 7?
- Show that if a node in a binary search tree has two children, then its successor has no left child and its predecessor has no right child.
- Demonstrate the insertion of keys 36, 27, 15, 47, 82, 33 into a hash table with collisions resolved by quadratic probing. Let the table have 11 slots (indexed 0 to 10), and let the auxiliary hash function be $h'(k) = k$ with $c_1 = 1$ and $c_2 = 3$.
- Show the red-black trees that result after successively inserting the keys 9, 12, 11, 17, 29, 33, 43 into an initially empty ~~binary search tree~~.

red black tree

2. (20%) A red-black tree is a balanced binary tree that has height $O(\log n)$.

- Show that a red-black tree with n internal nodes has height $O(\log n)$.
- What is the largest possible number of internal nodes in a red-black tree with black height k ? What is the smallest possible number?

高度，從點出發

claim 1, 點 $2^{bh(x)-1}$

$2^{2k} - 1$

$2^k - 1$

要推導

3. (20%) Consider a modification of the rod-cutting problem in which, in addition to a price p_i for each rod, each cut incurs a fixed cost of c . The revenue associated with a solution is now the sum of the prices of the pieces minus the costs of making the cuts. Give a dynamic-programming algorithm to solve this modified problem.

- Suppose that each cut incurs a fixed cost of 3. Determine an optimal solution to cut a rod of length 7 with the following table of prices p_i for length $i = 1, 2, \dots, 7$.

length i	1	2	3	4	5	6	7
price p_i	2	4	7	8	11	12	12

賺多少錢
扣成本

- Give a dynamic-programming algorithm to solve this modified problem.

相與數字

4. (15%) Describe an $O(n)$ -time algorithm that, given a set S of n distinct numbers and a positive integer $k \leq n$, finds the k elements in S whose values are closest to the median of S (excluding the median itself).

最接近中位數字的 k 個

Hint: The selected elements need not be symmetrically distributed around the median in the sorted order. For example, given 1, 2, 3, 8, 10, the two elements closest to the median 3 are 1 and 2 (excluding the median), both of which lie on the same side. This is only an illustration; do not assume the input is sorted.

元素
且不含
中位數
本人!

$O(1)$

Not NC

Greedy

5. (15%) In this question, you are given a set $C = \{1, 5, 10, 25\}$ of coin denominations, that is, pennies, nickels, dimes, and quarters. Your goal is to design an algorithm which, for any input value V in cents, outputs the minimum number of coins with denominations in C whose total value is exactly V . For example, to represent the value $V = 42$, we would choose one quarter, one dime, one nickel, and two pennies, a total of 5 coins. Your algorithm should take $O(|C|) = O(1)$ time.

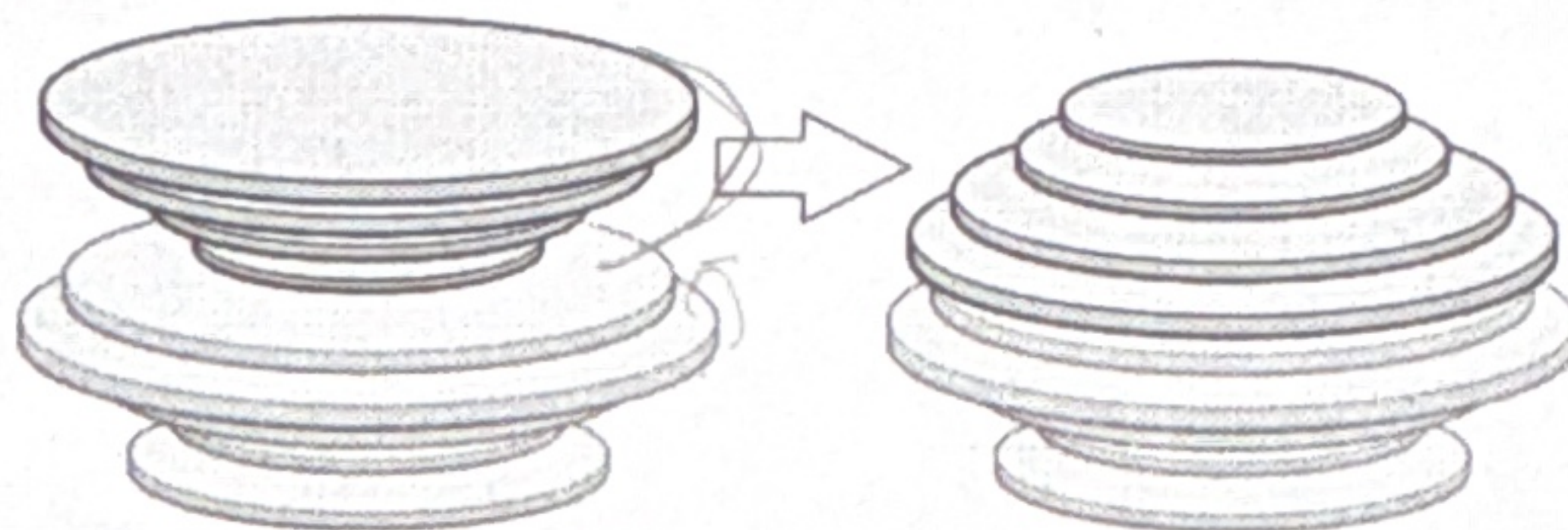
- (a) Describe your algorithm, argue briefly that it does indeed achieve the desired running time.
(b) Prove formally that it outputs the minimum number of coins needed to represent a given value V , for any value V it is given as input.

pancake $\frac{15}{2}$

6. (15%) Suppose you are given a stack of n pancakes of different sizes. You want to sort the pancakes so that smaller pancakes are on top of larger pancakes. The only operation you can perform is a flip-insert a spatula under the top of k pancakes, for some integer k between 1 and n , and flip them all over.

翻到 k 个 pancake
翻面

→ 唯一操作



- (a) Describe an algorithm to sort an arbitrary stack of n pancakes using as few flips as possible.
(b) Exactly how many flips does your algorithm perform in the worst case?

$$42 \div 25 = 1 \dots 17$$

$$17 \div 10 = 1 \dots 7$$

$$7 \div 5 = 1 \dots 2$$

$$2$$

$$\begin{array}{r} 1 \\ 25 \overline{) 42} \\ \underline{25} \\ 17 \end{array}$$

$$\begin{array}{r} 34 \\ 843 \overline{) 921} \\ \underline{843} \\ 78 \end{array}$$

$$\begin{array}{r} 129 \\ 848 \overline{) 129} \\ \underline{848} \\ 441 \end{array}$$

$$\rightarrow 1293484$$

$$\begin{array}{r} 78 \ 19 \ 24 \ 37 \\ 83 \ 91 \ 42 \ 73 \end{array}$$

$$84291$$

$$248192$$

$$241893$$

$$214894$$

$$124895$$